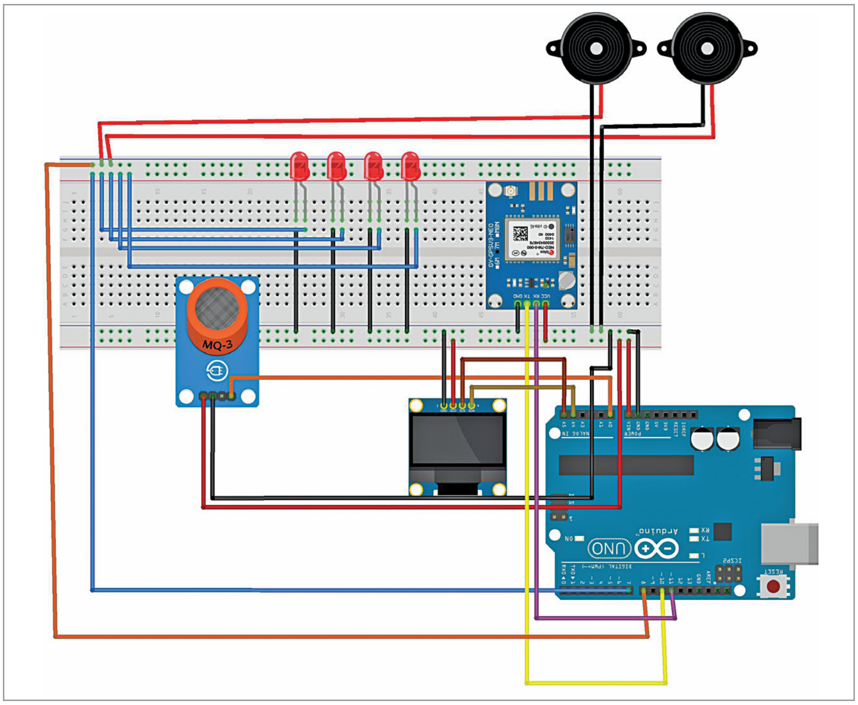
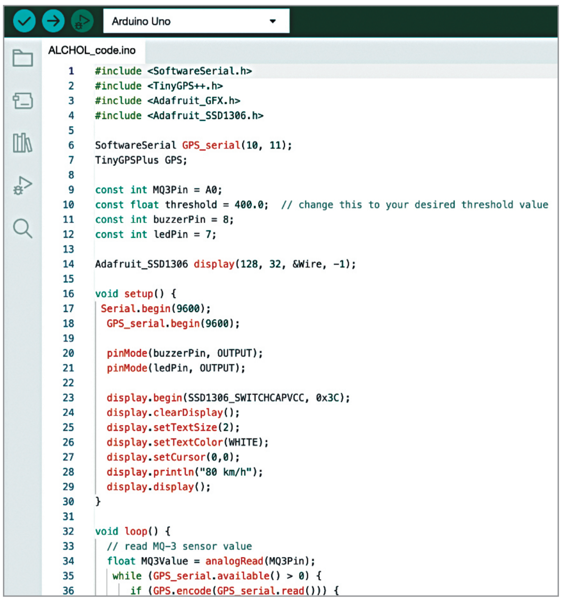


Fig. 3: Wiring/connection on breadboard



Arduino Uno

ALCHOL_code.ino

```
#include <SoftwareSerial.h>
#include <TinyGPS++.h>
#include <Adafruit_GFX.h>
#include <Adafruit_SSD1306.h>

SoftwareSerial GPS_serial(10, 11);
TinyGPSPlus GPS;

const int MQ3Pin = A0;
const float threshold = 400.0;  // change this to your desired threshold value
const int buzzerPin = 8;
const int ledPin = 7;

Adafruit_SSD1306 display(128, 32, &Wire, -1);

void setup() {
 Serial.begin(9600);
  GPS_serial.begin(9600);

  pinMode(buzzerPin, OUTPUT);
  pinMode(ledPin, OUTPUT);

  display.begin(SSD1306_SWITCHCAPVCC, 0x3C);
  display.clearDisplay();
  display.setTextSize(2);
  display.setTextColor(WHITE);
  display.setCursor(0,0);
  display.println("80 km/h");
  display.display();
}

void loop() {
  // read MQ-3 sensor value
  float MQ3Value = analogRead(MQ3Pin);
   while (GPS_serial.available() > 0) {
      if (GPS.encode(GPS_serial.read())) {
```

Fig. 4: Screen shot of source code

from alcohol impairment. Thus, this project is designed with the intention of preventing such incidents.

At the heart of this project lies an easily readable OLED display, with the Arduino functioning as a speedometer. Additionally, LED indicators and buzzers serve as intuitive communication tools. When alcohol impairment is detected by the highly sensitive MQ3 alcohol sensor, and the alcohol level exceeds the threshold value, the LEDs flash, and the buzzer emits distinct sounds. This alert persists until the reset button is pressed. This ensures that the driver acknowledges his or her impaired state and takes the necessary corrective action.

Furthermore, an optional speedometer can be incorporated into the system to monitor the vehicle's speed. By default, the code displays a random speed.

Additionally, a Neo 6M GPS module is utilised, offering the potential for real-time tracking of the vehicle's location. This expanded capability allows a Raspberry Pi or any laptop to be connected to the system. Upon detecting alcohol impairment, the project sends an email to a pre-determined address, with precise location information obtained from the GPS module.

Fig. 1 shows the author's prototype. Components required in this project are listed in the bill of materials Table.

Fig. 2 shows circuit diagram of the alcohol detector. The circuit includes an Arduino Uno board (MOD1), four LEDs (L1 through L4), two buzzers (B1 through B2), an OLED SSD1306 module (MOD2), a NEO 6M GPS module (S1), an MQ3 sensor (S2), and various other components. The wiring diagram is illustrated in Fig. 3.

Software

The project's software is developed using the Arduino IDE. To begin, the required libraries must be installed, namely 'TinyGPS + + .h', 'Adafruit_GFX.h', and 'Adafruit_SSD1306.h'. These libraries are initialised within the code. The alcohol detection function is implemented within the loop function. By default, a random speed limit is set for demonstration purposes, but this can be adjusted to match any speedometer. Once the code is ready, upload it by selecting the appropriate COM port and board.

Fig. 4 shows a screen shot of the source code.

For those interested in extending the project to include email alerts upon detecting alcohol impairment, Python is employed. The required libraries include 'serial' and 'smtp'. The Python code monitors the serial port for a message indicating high alcohol values. When such a message is received, an email is sent to the address specified in the code.

For the email of sender and password of sender and the receiver's email needed to be set in the code refer to Fig. 5.